

# KASRA AHMADI

Ph.D. Candidate, Computer Science and Engineering — ML Security & Post-Quantum Cryptography

Sunnyvale, CA | kasra.research@gmail.com | Google Scholar | GitHub | LinkedIn |

EB2-NIW Approved (U.S. Permanent Residency in Process)

## SUMMARY

Ph.D. candidate specializing in **Machine Learning Security** and **Fault-Resistant Post-Quantum Cryptography (PQC)**. First author of 10+ peer-reviewed publications, including a Best Paper Award at IEEE S&P 2025 Workshops. Experienced designing secure AI systems, cloud-native ML platforms, and production ML engineering workflows. Strong background in RAG security, federated learning privacy, and hardware fault/side-channel resilience for AI and cryptographic systems.

## EDUCATION

**Ph.D. in Computer Science** — University of South Florida, US

Jan 2022 – Oct 2026

CGPA: 3.93/4

**MSc in Information Technology Engineering** — Amirkabir University of Technology, Iran

Sep 2018 – Aug 2020

## WORK EXPERIENCE

**ML Research Engineer Intern** — RealmLabs

May 2026 – Present

Sunnyvale, CA

- Developed PII privacy and security workflows using frozen Qwen 3.5 and Llama 3.1 models, achieving a 3% F1 score improvement over ModernBERT by disabling causal masking to capture full bidirectional context.
- Building token-level PII probe models on decoder hidden states and integrating vLLM with sparse autoencoders (SAEs) for single-pass generation signal capture.
- Working on cost optimization and model routing for Claude Code by classifying session intent and routing to the appropriate Claude model.

**Graduate Research Assistant** — University of South Florida

Jan 2022 – Present

Tampa, FL, US

- Designed a human-in-the-loop, differentially private federated learning framework for fine-tuning LLMs; **won Best Paper Award at IEEE S&P 2025 Workshops**.
- Designed algorithm-level fault-injection and side-channel error-detection schemes for post-quantum cryptosystems (Kyber, Dilithium, ECC/ECSM) and DNN inference (MAED), benchmarked on FPGAs and ARM.
- Authored 10+ peer-reviewed papers; peer reviewer for IEEE TVLSI, TCAS-I, and ACM TECS.

**AI Engineer Intern** — TD Synnex

May 2025 – Aug 2025

Clearwater, FL, US

- Designed scalable, cloud-native AI platforms using LLMs, RAG, and multi-agent orchestration, with safety/security guardrails for production deployment.

**Software Engineer Intern** — AgWise (TransparencyWise)

May 2024 – Aug 2024

St. Petersburg, FL, US

- Built an AWS-based ETL pipeline (Lambda, Glue, S3) to process laboratory nutrient data for crop-recommendation models supporting up to 1M users.

## TECHNICAL SKILLS

**Languages:** Python, C++, C, Linux/Bash

**ML & LLMs:** PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LoRA, DP-SGD, Flower, vLLM, sparse autoencoders, RAG

**Security & Privacy:** PII detection, privacy-preserving ML, federated learning, differential privacy, adversarial ML evaluation, fault-injection analysis

**ML Systems:** ONNX, Docker, AWS (Lambda, Glue, S3, Solutions Architect), model routing, evaluation pipelines

## SELECTED PUBLICATIONS

**An Interactive Framework for Implementing Privacy-Preserving Federated Learning: Experiments on Large Language Models.** Kasra Ahmadi, et al. *IEEE Security and Privacy Workshops (SPW), 2025. Best Paper Award.* [arXiv](#) [GitHub](#)

**Efficient Algorithm Level Error Detection for Number-Theoretic Transform Assessed on FPGAs.** Kasra Ahmadi, Saeed Aghapour, Mehran Mozaffari Kermani, Reza Azarderakhsh. *ACM Transactions on Embedded Computing Systems, 2025.* [arXiv](#) [GitHub](#)

**MAED: Mathematical Activation Error Detection for Mitigating Physical Fault Attacks in DNN Inference.** Kasra Ahmadi. *In submission.* [GitHub](#)

## SELECTED PROJECTS

**PII-360** — [GitHub](#) · [Product](#)

Open-source Chrome extension that detects personally identifiable information (PII) in images and PDFs using ONNX-based models running entirely on the client's local machine.

## CERTIFICATIONS

AWS Certified Solutions Architect - Associate • LangChain for LLM Application Development • Deep Neural Networks with PyTorch (IBM) • ETL and Data Pipelines with Shell, Airflow and Kafka (IBM) • Attention Mechanisms and Transformer Models • Generative AI - Technical AI Advisor (Nvidia) • Fundamentals of LLMs (HuggingFace) • Fine-tuning Language Models (HuggingFace)